

MRIDUL KUMAR

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EDUCATION

Indian Institute of Technology, Patna
B.Tech in Artificial Intelligence and Data Science

NOV 2022 – MAY 2026
CPI: 8.54

EXPERIENCE

RapidFort Inc.
Software Engineer (Docker, Kubernetes, Helm, Python, Bash, Linux, SBOM)

JAN 2026 – PRESENT
Bangalore, India

- Engineered near-zero-CVE container images by stripping unused packages and dependencies, reducing attack surface by up to **95%** for cloud-native production workloads.
- Validated hardened images on live **Kubernetes clusters**, writing Python and Bash test automation to catch regressions before release across the curated image catalog.
- Maintained multiple open-source image repositories compliant with **STIG/FIPS** standards, supporting customer certification requirements under **FedRAMP, CMMC, SOC 2, and NIS 2**.
- Delivered continuous vulnerability remediation and image updates, consistently meeting security SLAs for enterprise and government customers.

SIEMENS
Software Developer Intern (Python, Flutter, NodeJS, MQTT, OPC-UA, FastMCP)

MAY 2025 – AUG 2025
Bangalore, India

- Engineered an HTTP-exposed **MCP** client using **FastAPI** and **FastMCP** to orchestrate servers controlling robotic arms, autonomous vehicles, and inventory systems.
- Engineered a real-time **SSE** dashboard for live telemetry and remote MCP control, powered by low-latency **OPC-UA** and **MQTT** data pipelines across industrial automation systems.
- Developed a mobile application that digitizes static PDF workflows into interactive, step-by-step guides, reducing operational labor overhead by nearly 50%.

SSquad Ventures

Full Stack Developer Intern (Flutter, Dart, Express.js, MongoDB, RESTful APIs, Git, Postman) Remote/India

JAN 2025 – APR 2025

- Built a social commerce **Outgoingg** platform with Flutter and Node.js/Express, reducing API latency by 35% via query indexing, caching, and pagination.
- Implemented best practices in full-stack development including **state management, modular architecture, secure token handling, and CI/CD workflows** for maintainable and high-performing releases.

PROJECTS

Hypervisor Based Malware Defense System | Powershell, HyperDbg, Python, VM, Intel VT-x **GitHub**

- Built a stealthy hypervisor-based malware analysis engine at Ring -1 (VMX root) via HyperDbg, intercepting 24 Windows kernel syscalls through live kernel structure traversal and EPROCESS side-channels — fully invisible to guest-level detection.
- Automated a 12-step, stateless detonation pipeline (VM reversion, FakeNet simulation, named-pipe sync) for unattended dynamic analysis and zero-footprint, in-VM PE execution across 909 samples.
- Designed a dual-modality ML classifier fusing sequential syscall embeddings with process lineage graphs to detect 7 attack patterns mapped to **MITRE ATT&CK** (Process Hollowing, APC Injection, DLL Sideload), reaching **0.751 AUROC**.

Multi-threaded Proxy server | C, Multithreading & Sockets, TCP/IP

GitHub

- Architected a multithreaded HTTP proxy server in C, efficiently managing 400+ simultaneous TCP connections using POSIX threads, semaphores, and non-blocking I/O.
- Implemented full HTTP/1.1 compliance with persistent connections, CONNECT tunneling, and intelligent header rewriting for seamless client-server communication.

SKILLS & COURSEWORK

Languages: C/C++, GoLang, Python, Dart, JavaScript, TypeScript, Bash

Frameworks & Libraries: FastAPI, Flutter, React.js, Express.js, Tailwind CSS

DevOps & Tools: Docker, Kubernetes, Git/GitHub, Postman, VS Code, VirtualBox, VMware Workstation , Linux/Ubuntu, DRAKVUF, CAPEv2, HyperDbg

Cloud & Databases: Firebase, MongoDB, MySQL, SQLite

Systems & Networking: Model Context Protocol, Multithreading, TCP/IP, UDP, Socket Programming, Wireshark, tcpdump, CVE Remediation, SBOM Generation, Container Hardening, Vulnerability Assessment

API & Communication: REST APIs, WebSockets, SSE (Server-Sent Events), MQTT, OPC-UA

ACHIEVEMENTS

- Specialist on Codeforces** with peak rating of 1433.

Profile